# Getting Started with Multipass: Installation and Basic Usage Guide

---

## On Windows

### Option 1: Using Winget (Recommended)

```
winget install Canonical.Multipass
```

### Option 2: Using the Web Installer

1. Visit the Multipass Installation Page.
2. Download **Multipass for Windows**.
3. Run the installer.
4. Select your virtualization backend:

- **Hyper-V** if you are on Windows 10/11 Pro, Enterprise, or Education (or select Hyper-V for Multipass 1.17+).
- **Oracle VM VirtualBox** if you are on Windows 10/11 Home.

---

## On macOS

### Option 1: Using Homebrew (Recommended)

```
brew install --cask multipass
```

### Option 2: Using the Web Installer

1. Visit the Multipass Installation Page.
2. Download **Multipass for macOS**.
3. Run the installer.

---

## Create and Start a Basic Linux System

See what Linux images are available:

```
multipass find
```

Launch a specific version (e.g., Ubuntu 24.04):

```
multipass launch 24.04 --name my-2404-ubuntu
```

*Alternatively, using an alias:*

```
multipass launch noble --name my-2404-ubuntu
```

Run the following command in your terminal to create and start a fresh Ubuntu Linux instance (Multipass automatically pulls the latest stable Ubuntu LTS by default):

```
multipass launch --name my-linux-vm
```

### Customizing Resource Allocation (Optional)

By default, Multipass allocates a VM 1 CPU core, 1 GB of RAM, and 5 GB of disk space. You can customize these resources during launch:

```
multipass launch --name custom-vm -c 2 -m 2G -d 10G
```

- `-c 2` : Allocate 2 CPU cores
- `-m 2G` : Allocate 2 GB of memory
- `-d 10G` : Allocate 10 GB of disk space

---

## Managing and Using Your Virtual Machine

### Check Running Instances

```
multipass list
```

### Mount Local Storage

```
multipass mount C:\monal\FDE my-linux-vm:/home/ubuntu/share
```

### Enter the Linux Shell

```
multipass shell my-linux-vm
```

### Navigate to the Shared Folder and Verify

```
cd ~/share
# Or alternatively: cd /home/ubuntu/share

ls -la
```

### Exit the Linux Environment

Press `Ctrl + D` or type:

```
exit
```

### Permanently Remove the Storage Link

```
multipass unmount my-linux-vm
```

### Stop the VM

```
multipass stop my-linux-vm
```

### Start a Stopped VM

```
multipass start my-linux-vm
```

### Delete the VM

```
multipass delete my-linux-vm
```

### Permanently Remove Deleted VMs (Purge)

```
multipass purge
```

### Check Details About the Linux System

```
multipass info my-linux-vm
```